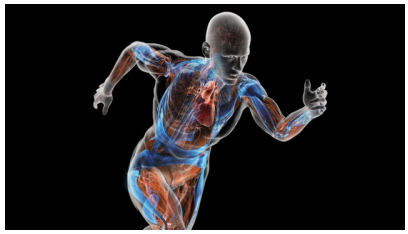
Lecture 17

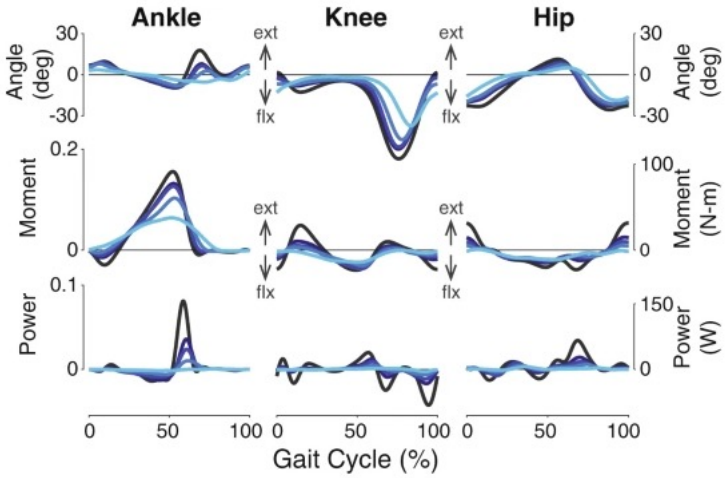
SM2501/FSG3084

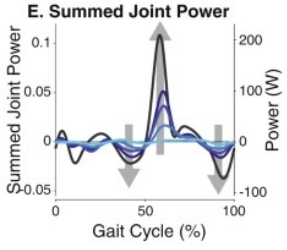


- Influence of walking speed
- Running
- A few ongoing research topics
- Summary

1

Effects of *very slow* speed in walking

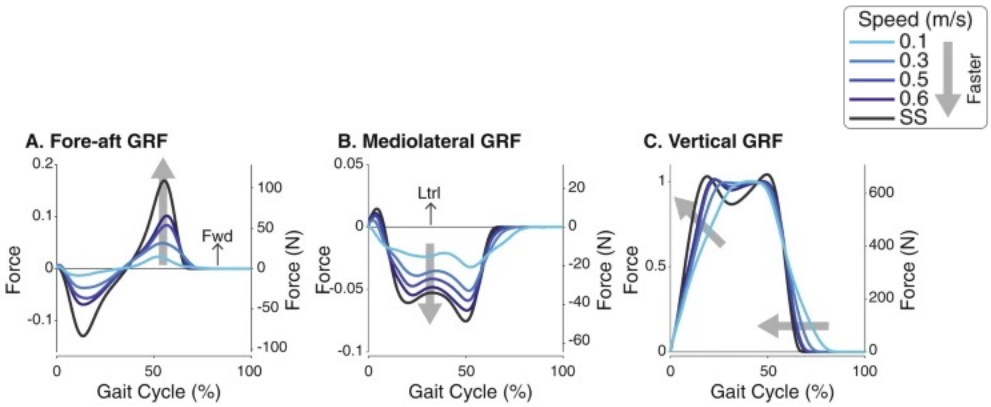




Wu et al, Sci Rep 9, 2019

2

Effects of *very slow* speed in walking

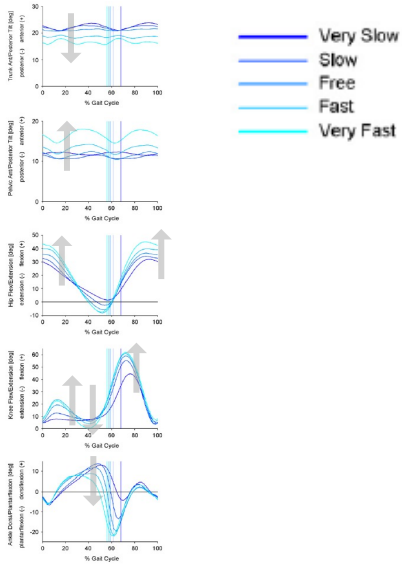


Wu et al, Sci Rep 9, 2019

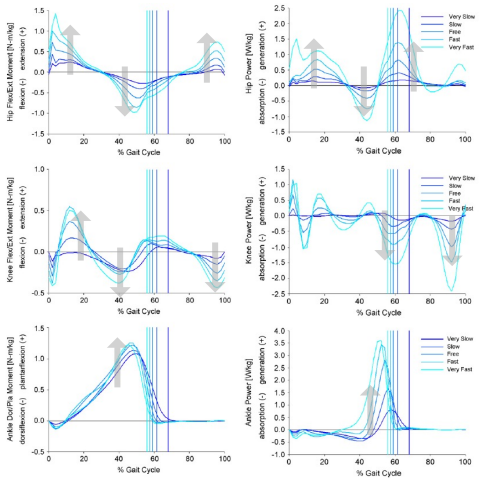
3

Gait over a range of walking speeds

Kinematics



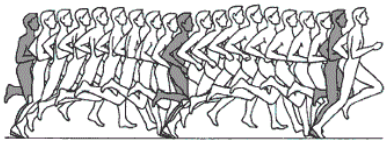
Kinetics



Schwartz et al, J Biomech 41, 2008

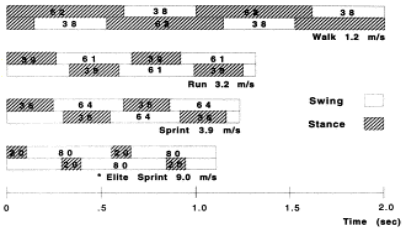
4

Running





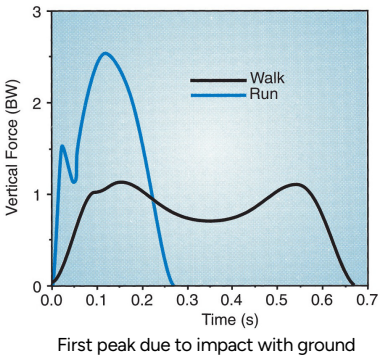
- Demarcated when double stance is replaced by 'double float''(Point A)
- Running vs. Sprinting – earlier demarcated by foot position at initial contact: changes from hindfoot to forefoot with increasing speed (Point B)
  - ~80% long distance runners strike with rearfoot, most of others with midfoot.



Novacheck, Gait Posture 7 1998

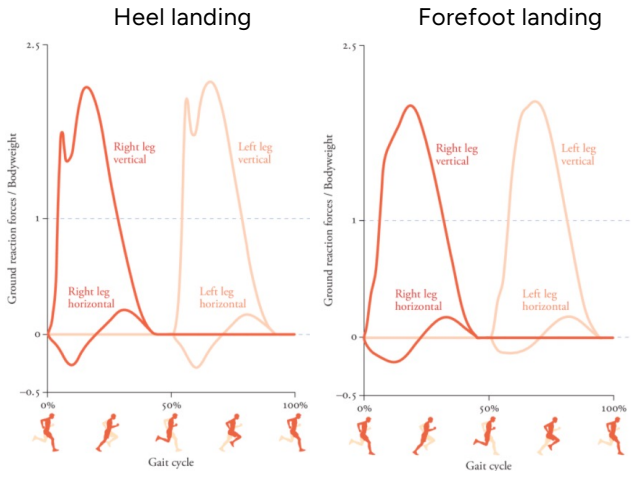
5

GRF



First peak due to impact with ground

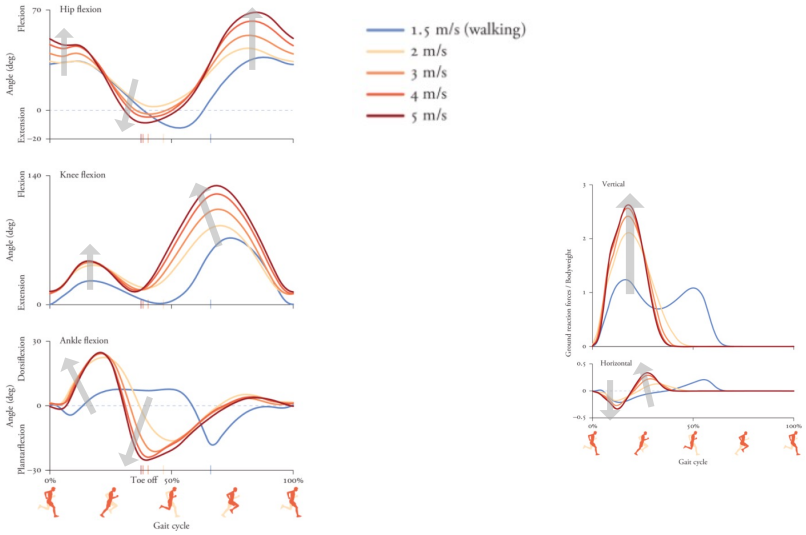
Hamill, Knutzen and Derrick 2015



Uchida and Delp 2020

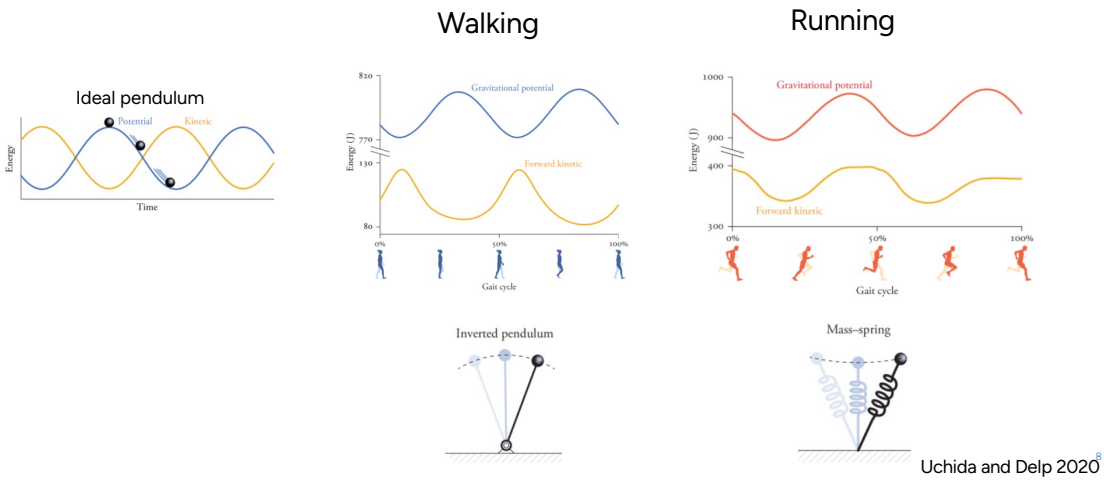
6

Kinematics and GRFs



7

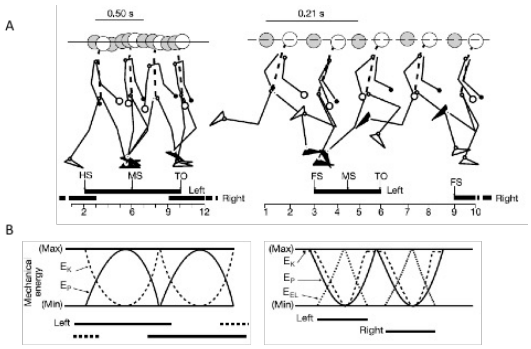
Mechanical efficiency



Uchida and Delp 2020[8]

8

Mechanical efficiency



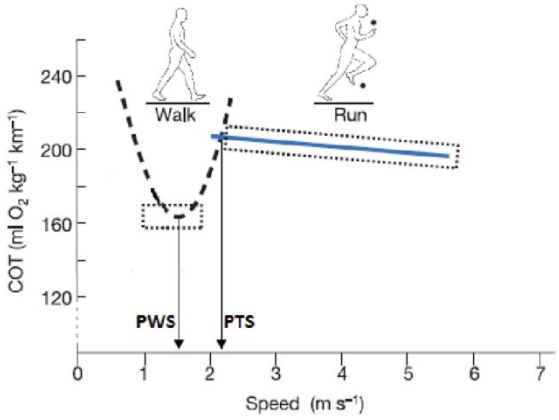
Potential and kinetic energy of COM in phase in running
Elastic strain energy stored and released in tendons and ligaments

Running efficiency by
- Storage and return of elastic potential energy in tendons
- Energy transfers in biarticular muscles
  - Hamstrings in late swing: transfer energy from moving tibia to pelvis to augment hip extension
  - Similarly, rectus femoris in early swing

Lazzer et al. ECOG Obesity eBook

9

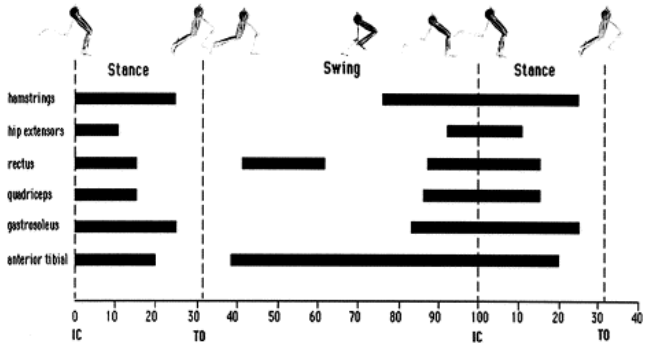
Metabolic cost



Sedentary Lifestyle, 2016

10

Muscle activity



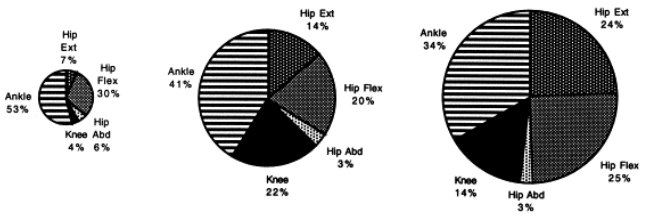
Generally, most activity just before and just after initial contact

High degree of co-contraction

Novacheck 1998

11

Contributions to positive work



Walk (1.2 m/sec) **Run** (3.2m/sec) **Sprint** (3.9 m/sec)

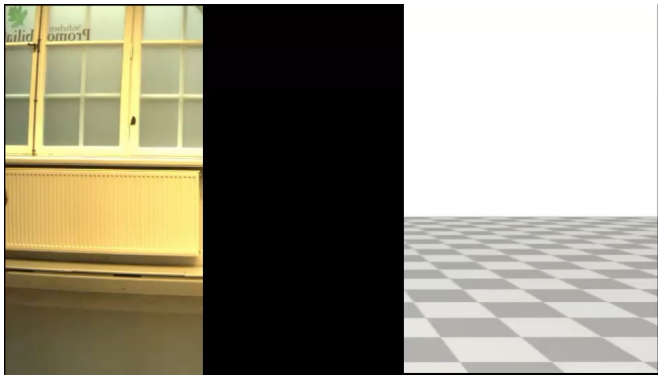
Novacheck 1998

12

Overview of current research

KTH MoveAbility

Decoding, predicting and enhancing human movement



Mission: to improve the quality of life in persons with motor disability by developing and implementing technologies that enable them to contribute beyond the best of their abilities

Research group at KTH MoveAbility



Faculty
Prof. Lanie Gutierrez-Farewik
Assoc. Prof. Ruoli Wang

Lab manager
Antea Destro

Post docs
Israel Luis
Liangliang Xiang
Jennifer Maier
Zhongzheng Wang

PhD students
Minh Truong
Ibrahim Hasan
Linghui Xu
Yuanyang Zhang
Zhihao Duan
Ruoyu Huang
Zhaoyuan Wan
Guillaume Flinois
Floor Janssen
Gustaf Sundell
Tyler Witzig

External PhD students
Tim Aeppli (MD, KI)
Alexandra Palmcrantz (KI)
Axel Fredriksen (KI)
Mariah Sabioni (KTH CBH)
Vera Fedosova (KI)

Visiting PhD student
Lianzheng Niu (Beijing)

Visiting MS students
Lorenzo Cortinovis (Univ. Padua)
Angelica Di Cillis (Univ. Torino)

Research Engineers
Jessica Cistakova
Lorenzo De Bei

foundation promobilia Vetenskapsrådet digital futures

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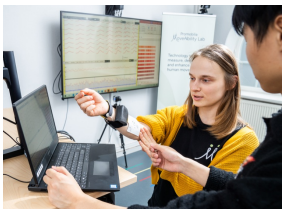
- 1.5 billion people worldwide live with a disability. This number is increasing.
- By 2050, world's 60+ population will double
- Musculoskeletal disorders -> sick leave
- Mobility -> overall health and quality of life
- In Sweden, 150,000 people use a wheelchair

➢ Increasing demand for technology to improve mobility

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Promobilia MoveAbility Lab research infrastructure







Motion analysis (10 Vicon V16 cameras)
4 fixed + *3 portable force plates (AMTI)
*IMUs (Myon aktos-t)
*EMG (Myon aktos-nano)
*Ultrasound (Philips)
*Pressure insoles (pedar, Moticon)
Instrumented treadmill (AMTI)
*High-density EMG 400 + 64 channels (OTB)
*Metabolic cost measurement (Cortex Metamax)
*Depth cameras

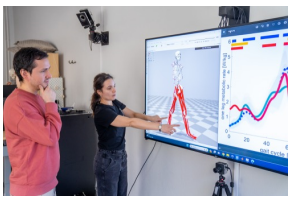
Simulation (OpenSIM and AnyBody)

* Modular/portable

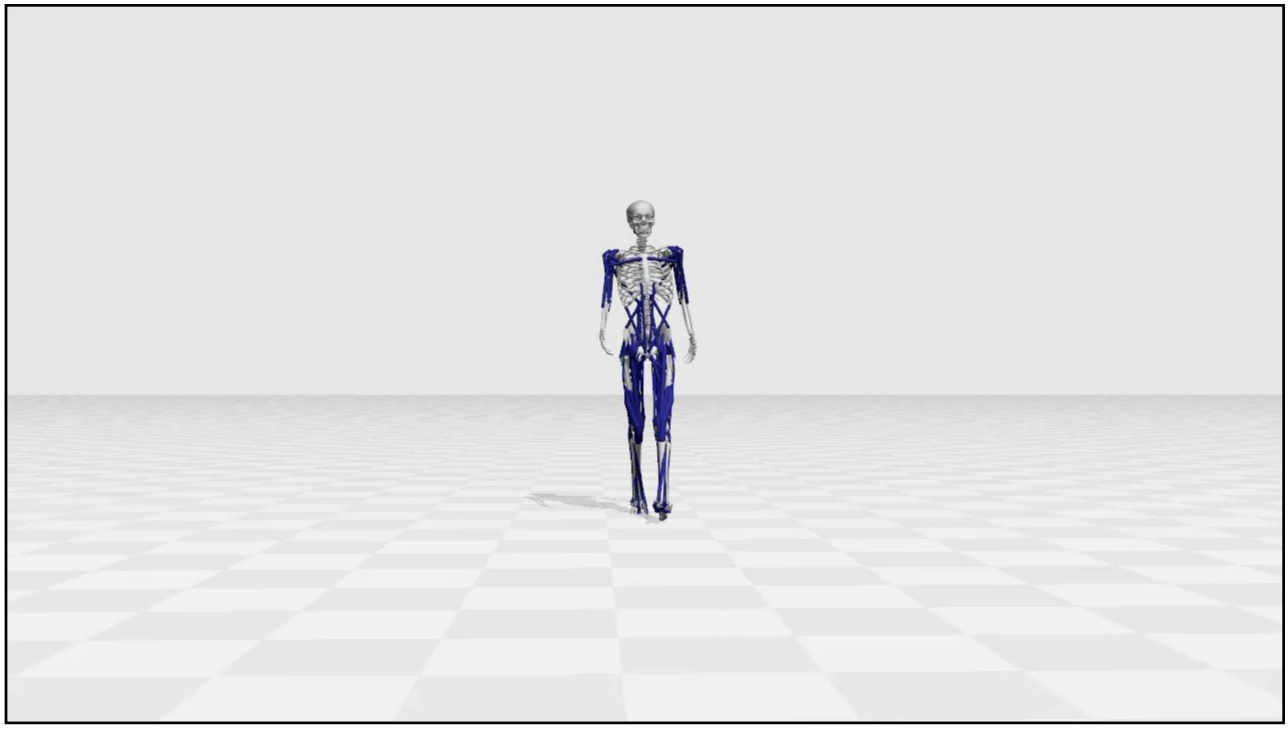




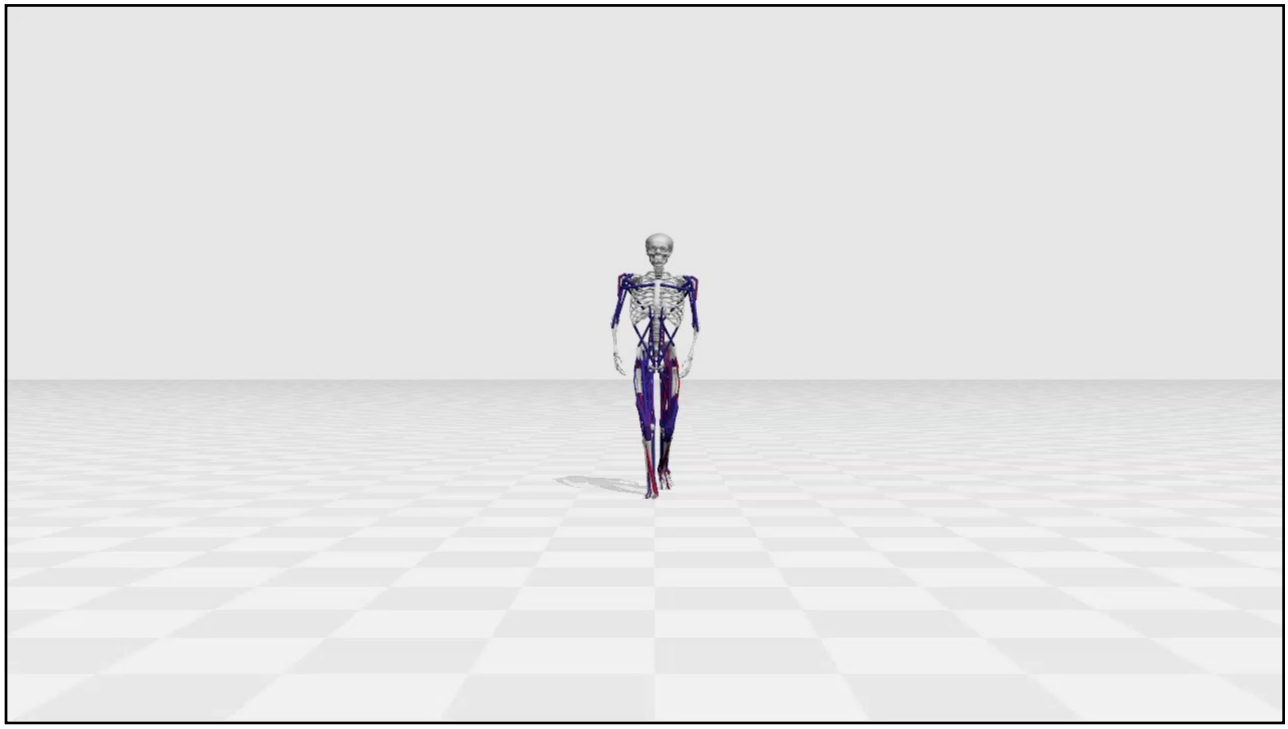




17



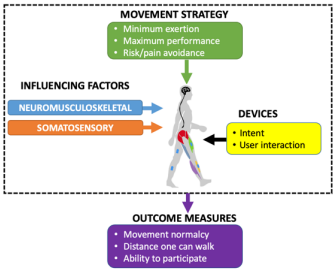
18



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Research Theme A: Decoding Movement



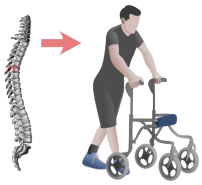


- Why do we walk like we do?
- What changes if we have pain? Loss of sensation? Balance disturbance?
- What changes occur in musculoskeletal tissues after a neurological injury?
- How does one (re)learn movement after an injury?
- How does the environment affect the way we move?
- What is a good outcome?

20

Movement performance <-> motor & sensory function
In persons with iSCI

Both are classified clinically as ASIA D





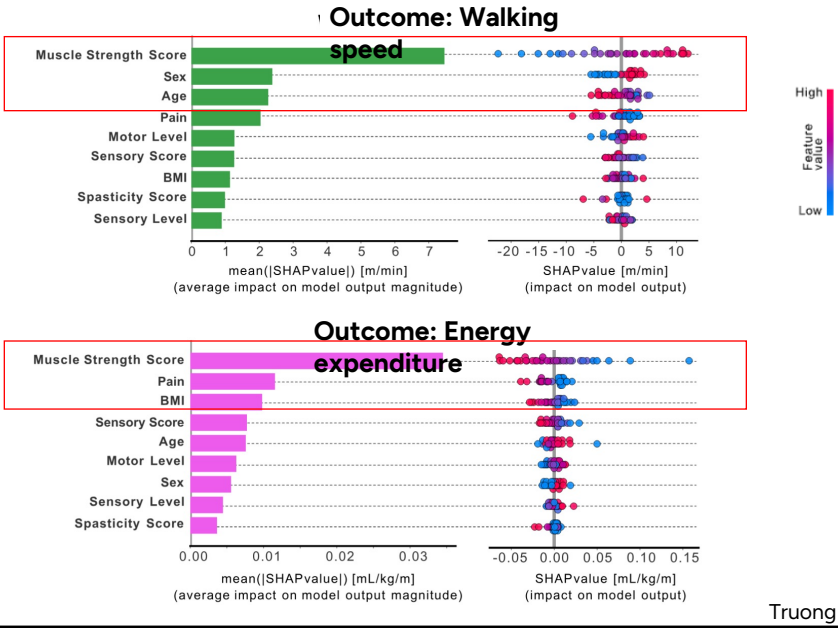


Goals:
- Quantify how motor and sensory function influence movement performance
- Constructive predictive data-driven models

Collaboration with Aleris Rehab Station/Spinalismottagningen & KI

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Group level



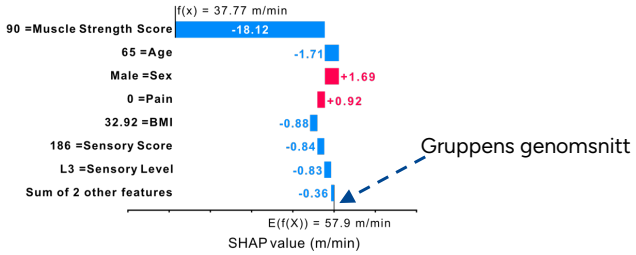
Truong et al., submitted 2025

22

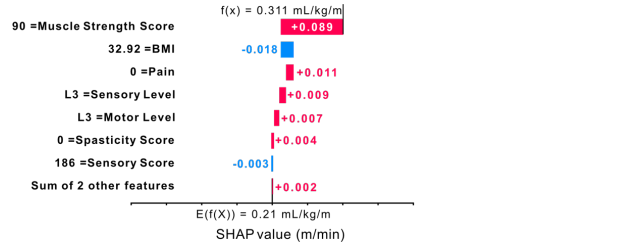
Individual level



Walking speed

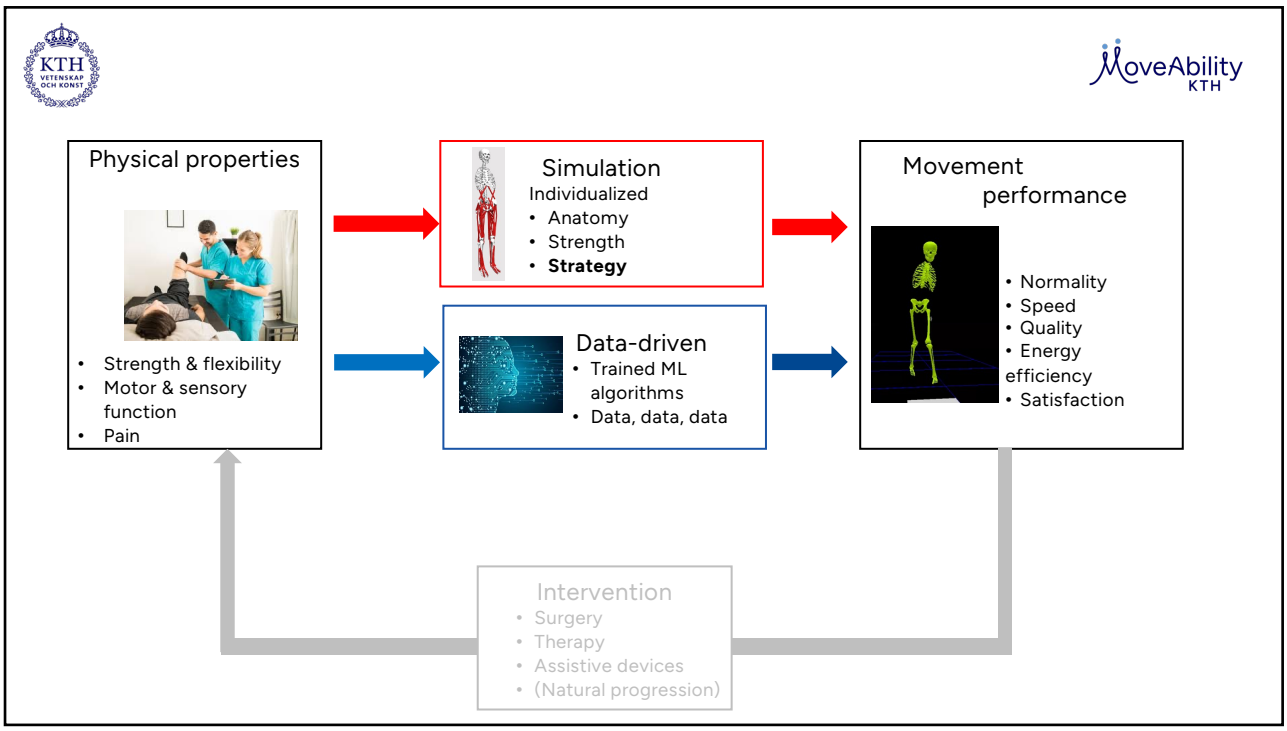


Energy expenditure



Effects: Tool for targetted intervention for max effect
Next steps: Populations: *Stroke, CP*, TBI, Parkinsons. Other performance outcomes

23



24

Research Theme B: Measurement and analysis movements

Movement analysis anywhere

Smartphone RGB +depth cameras



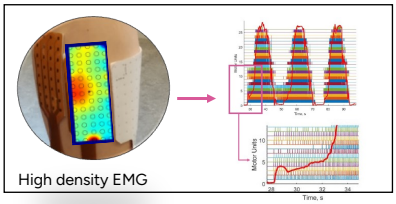


Augmented reality marker with web camera

Shoulder loading in wheelchair activities



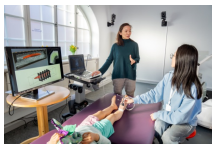
Neuromuscular function

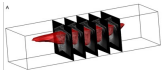


High density EMG

Muscle tissue and structure

3D freehand ultrasound
Ultrasound shear wave elastography
(MR-DTI)
(Magnetic resonance elastrography)



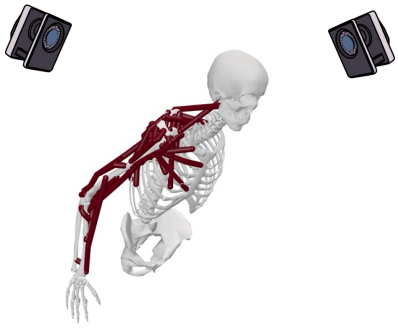


25

The wheelchair user's shoulder

44% of wheelchair users experience shoulder pain[1]





[1] Liampas et al. Musculoskeletal Pain Due to Wheelchair Use: A Systematic Review and Meta-Analysis. *Pain Ther.* 2021

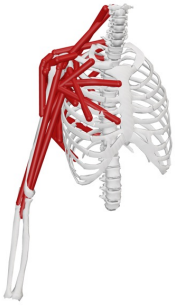
Collaboration with Aleris Rehab Station, Mid Sweden University and the Swedish School of Sport and Health Sciences

26

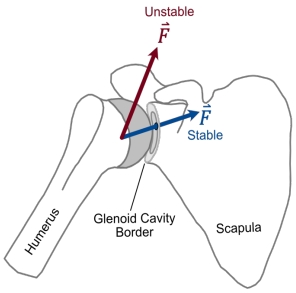
26

Shoulder Model: Contact Force

Model



Glenohumeral Stability [1]



In vivo measurements



orthoload.com

Realistic predictions of shoulder loading

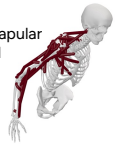
[1] Hasan I. M. I., et al. Modeling glenohumeral stability in musculoskeletal simulations: A validation study with in vivo contact forces. *bioRxiv*. 2025.

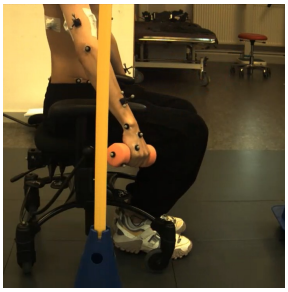
27

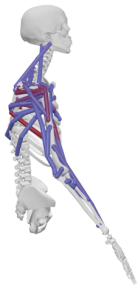
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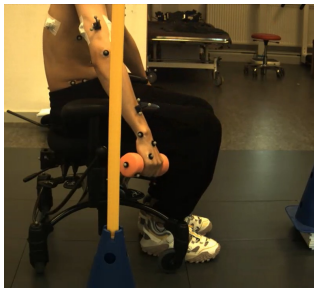
Shoulder loading with posture

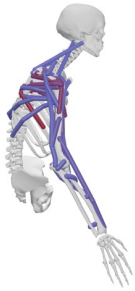
Spine-integrated thoracoscapular shoulder (**SITS**) model







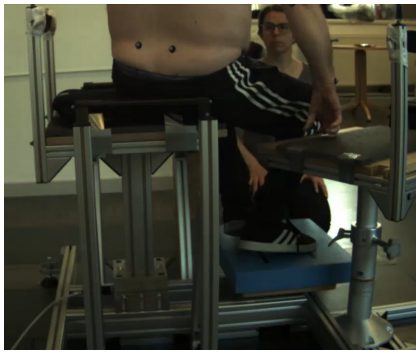


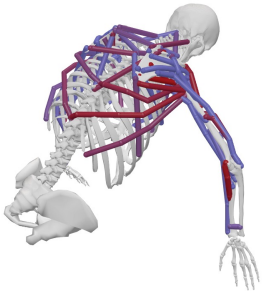


If slouching, shoulder loading as much as twice as if sitting upright

28

Shoulder loading in transfers



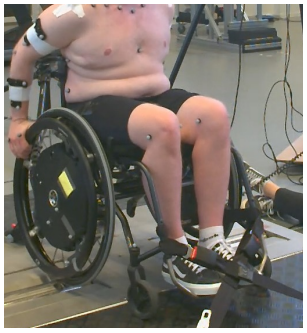


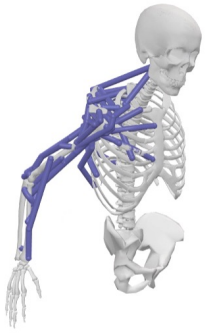
Shoulder loading as high as 10 times body weight!

29

29

Wheelchair propulsion





How do different propulsion techniques affect shoulder movement, ligament forces and muscle moment arms

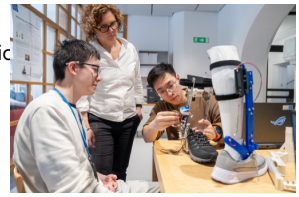
30

Research Theme C: Enhance human movement

Design & development of assistive exoskeletons

Strategy:
- To have genuine impact through goal-specific, comfortable and responsive devic
- Assistive or resistive as needed

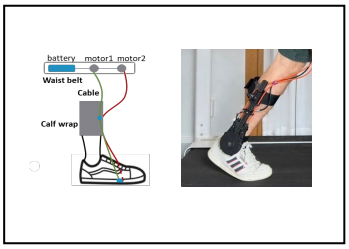
Approaches:
- Algorithms to detect of user's abilities and intended movements
- Simulation-based design and control
- Human-in-the-loop optimization



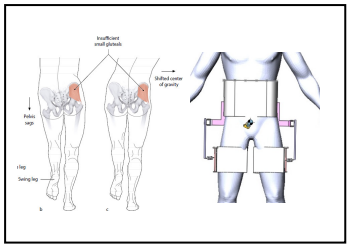
(Quasi)Passive ankle exoskeleton



Active ankle exoskeleton



Hip exoskeleton for abduction assistance



Effects: Individualized assistance for specific goals

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Background: Exoskeletons to reduce energy cost

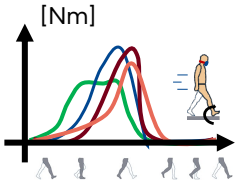
Experimental approach



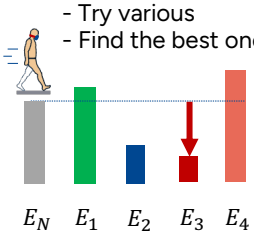
Simulation approach (complementary)



Moment [Nm]



- Try various
- Find the best one



- Identify (hypothetically) optimal assistance
- Less burden/iterations for participants

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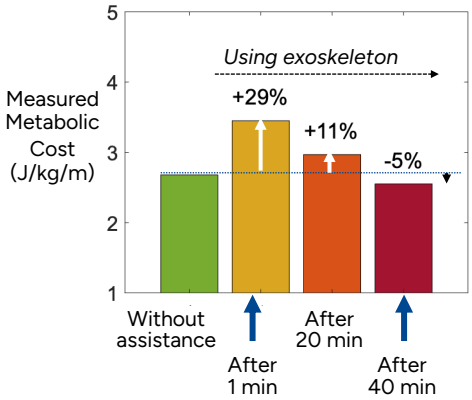
32

Adaptation to ankle exoskeleton during normal walking

Metamax 3b spiroergonometer

Biomotum SPARK ankle exoskeleton

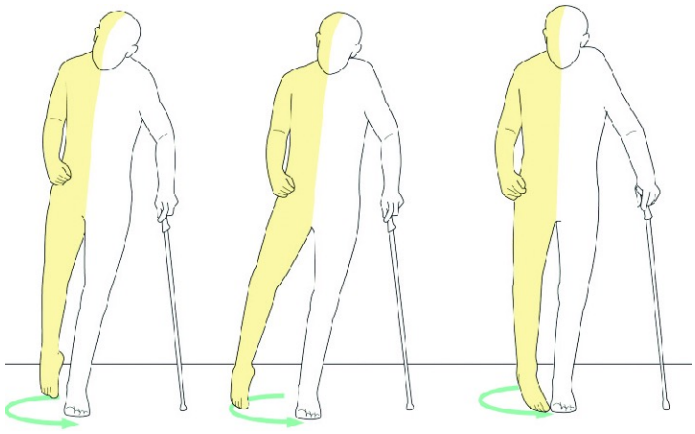




Findings: Adaptation can take minutes/hours (days?). Difficult to predict.

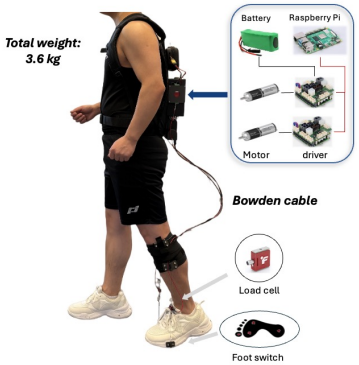
33

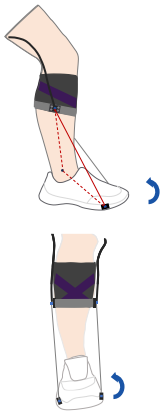
Dropfoot



34

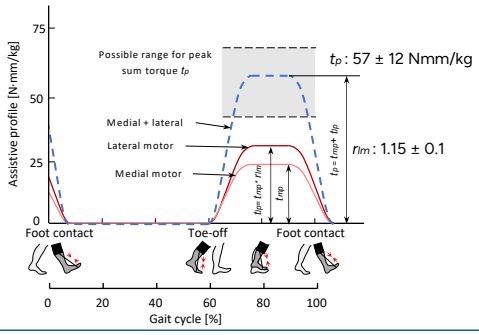
Design and control of ankle exoskeleton & biplanar assistance





Variables and constraints:

- Sum of the torque magnitude of two motors, t_p
- Torque ratio between lateral and medial motors, r_{lm}



Zhang et al. Front. Neurorob. 2024; Zhang et al. IEEE RA-L 2025.

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Participants and Methods

Participants

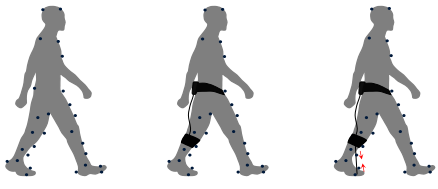
n=5, chronic phase after stroke

- 2M / 3F, height: 168 [159, 184] cm, weight: 93 [73, 100] kg, age: 46 [42, 58] yrs
- Time since stroke 39 [21, 60] months, Fugl Meyer- Lower Extremity 25 [21, 29].

Experimental protocol

Participants walked on the lab walkway at self-selected speeds in four sessions:

- **NoExo**: with standardized sneakers.
- **Unpowered Exo**: with identical sneakers, cables disconnected.
- **Familiarization**: with powered exoskeleton and different assistive profiles.
- **HIL**: Human-in-the-loop optimization to seek optimal assistive profiles.
  - All profiles subjectively **rated** 0-10 for perceived improvement



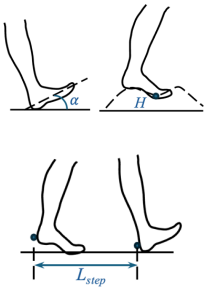
36

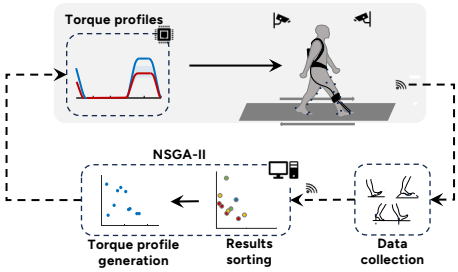
Methods: Multi-objective HIL optimization

Algorithm: Non-Dominated Sorting Genetic Algorithm (NSGA-II) was implemented into the multi-objective HIL framework.

Objective functions:

- Minimal foot kinematics (α, H) asymmetry
- Minimal step length asymmetry





Iterations: 3 generations × 10 torque profiles

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Results

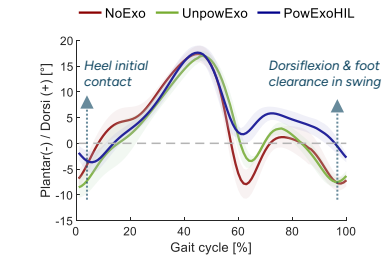
43 year old woman, post stroke in chronic phase

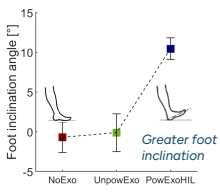


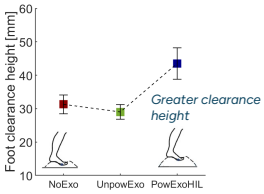
No exo



With exo







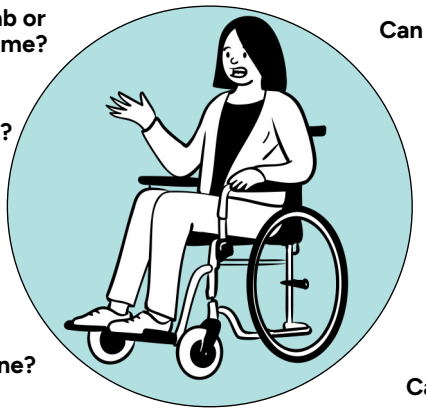
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User-Centered Design for Assistive Exoskeletons

How does it know how I'm going to move?

Is it just for rehab or can I use it at home?

Can I go to the bathroom while using it?



What do I do if it breaks?

Will it help me keep up with my kids?

Is it tracking me?

How loud is it?

How do I clean it?

Can it sense if I'm going to fall?

Can I take it on a bus? On a plane?

Can I put it on and take it off by myself? How long does that take?

Can this device realistically provide meaningful assistance?

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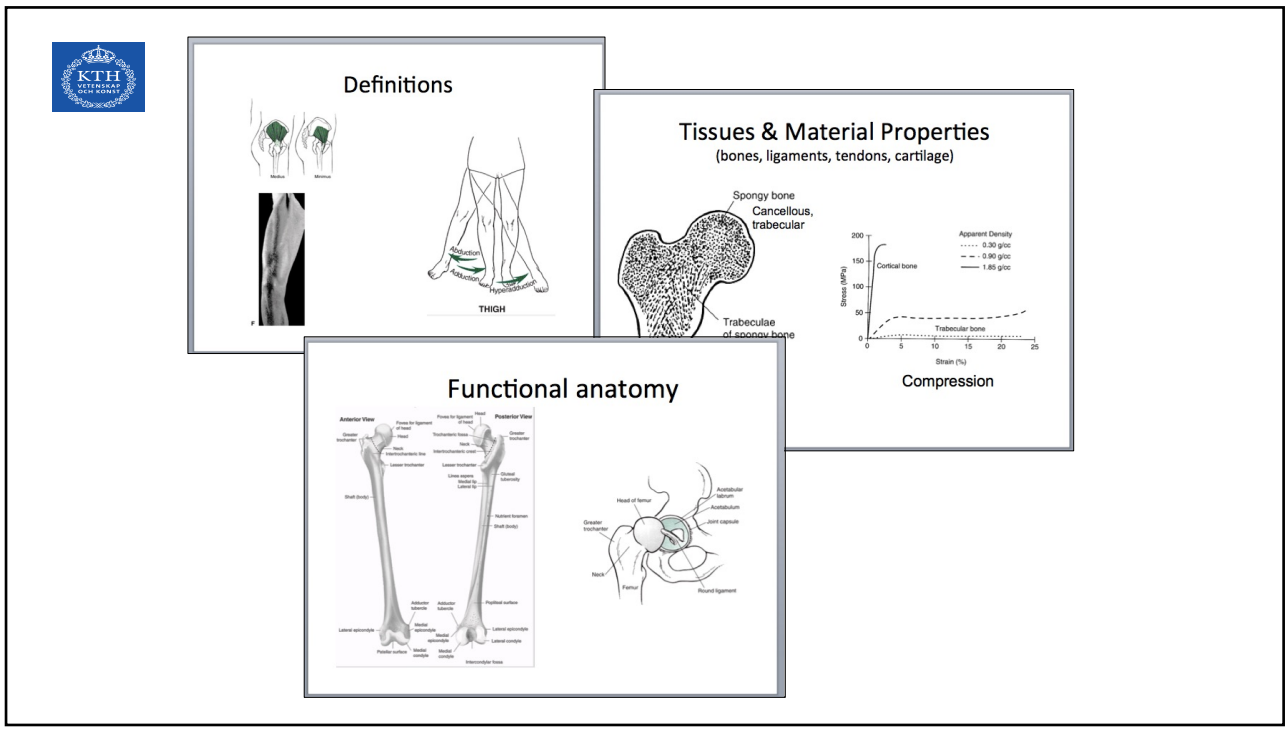
Course in review....

Objectives
This course is aimed towards upper level undergraduates and masters students.

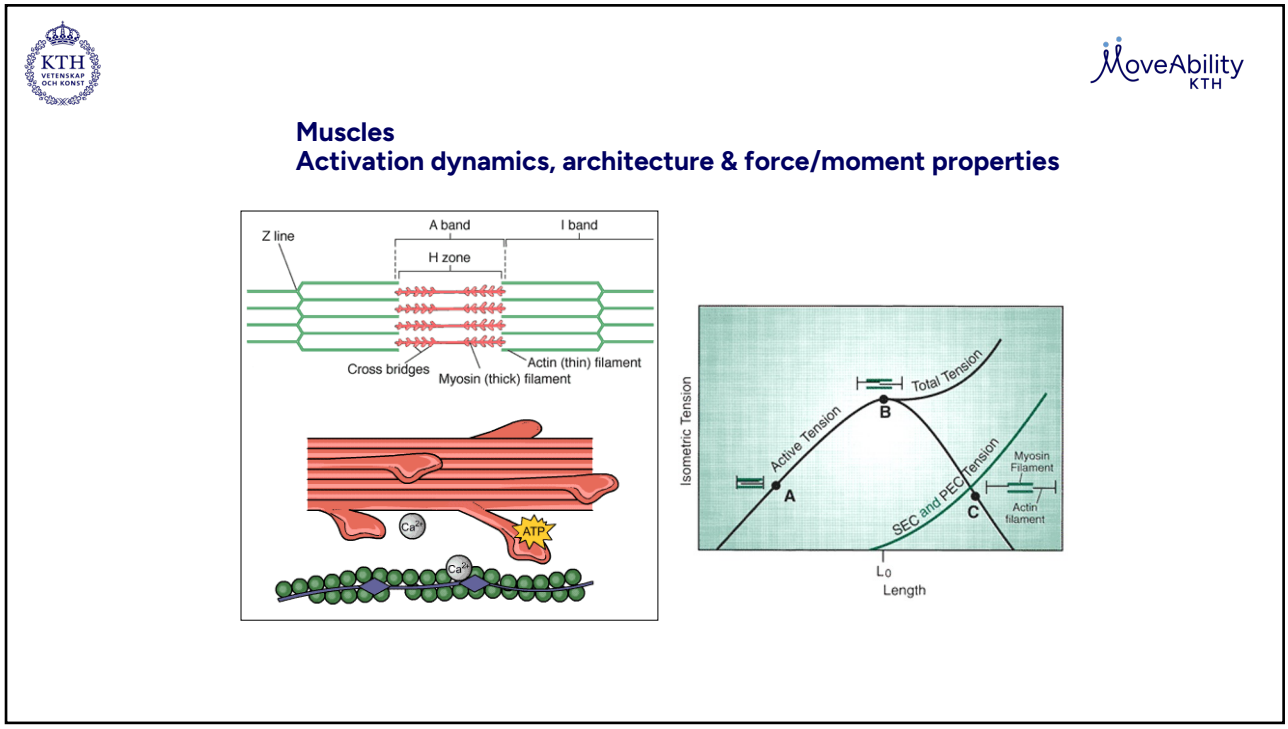
After the course, the student should be able to:

- Apply principles of classical mechanics to the study of human motion
- Describe motion with precise, well-defined mechanical and anatomical terminology
- Describe the internal and external forces acting on the body during typical human activities
- Understand how muscle actions control movements
- Understand the dynamic relationship between structure and function in the musculoskeletal system
- Model muscle activation and movement
- Understand the methods and limitations of different experimental and analytical techniques used

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Definitions
Abduction
Adduction
Hyperabduction
THIGH
Tissues & Material Properties
(bones, ligaments, tendons, cartilage)
Spongy bone
Cancellous, trabecular
Trabeculae of spongy bone
Cortical bone
Trabecular bone
Apparent Density
0.30 g/cc
0.90 g/cc
1.85 g/cc
Stress (MPa)
Strain (%)
Compression
Functional anatomy
Anterior View
Posterior View
Head of femur
Acetabular labrum
Acetabulum
Joint capsule
Greater trochanter
Neck
Femur
Round ligament

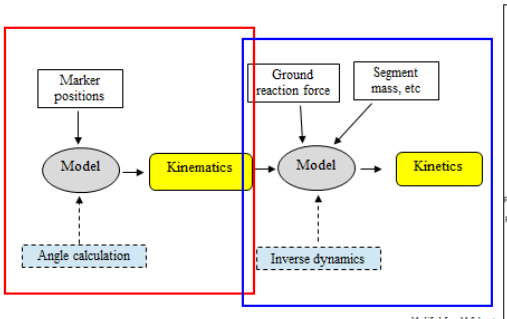
41

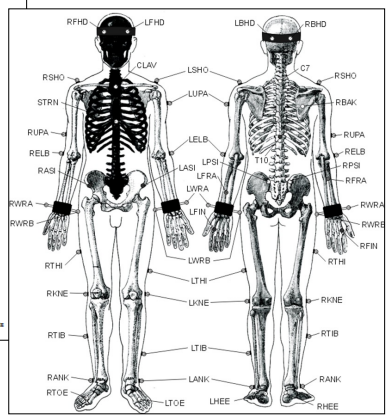

MoveAbility
KTH
Muscles
Activation dynamics, architecture & force/moment properties
Z line
A band
I band
H zone
Cross bridges
Actin (thin) filament
Myosin (thick) filament
Ca2+
ATP
Isometric Tension
Total Tension
Active Tension
SEC and PEC Tension
A
B
C
Myosin Filament
Actin filament
L0
Length

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Motion analysis
Theory and demonstrations

Experimental kinematics and kinetics



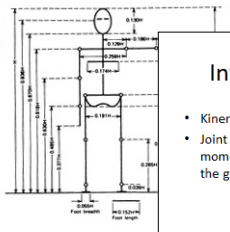


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Kinematics and kinetics

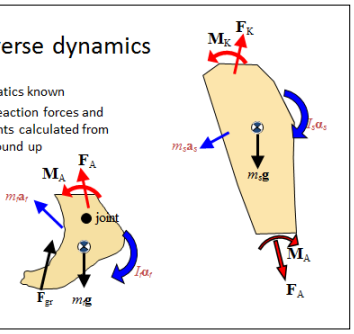
Anthropometry

– Ex. Body segment lengths as fraction of body height (Demster 1955, Drillis and Contini 1966)



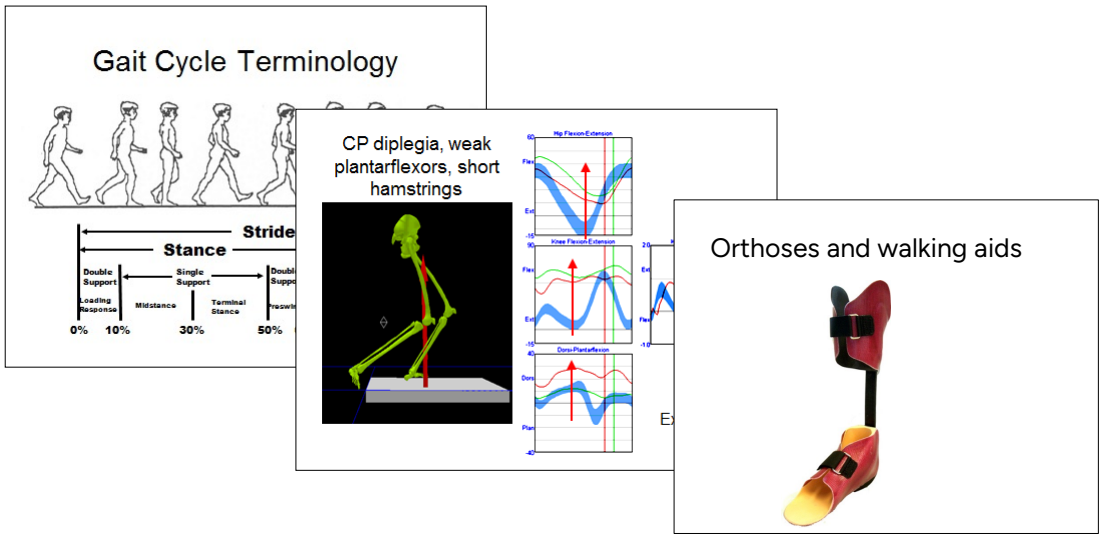
Inverse dynamics

- Kinematics known
- Joint reaction forces and moments calculated from the ground up



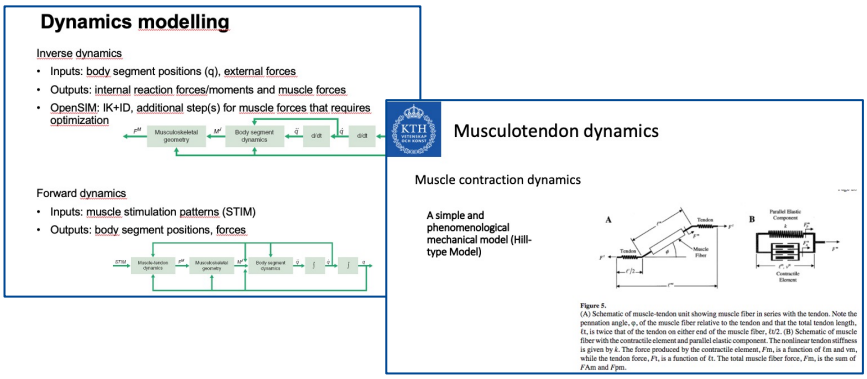
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Normal and pathological movement biomechanics



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Modelling and motor control



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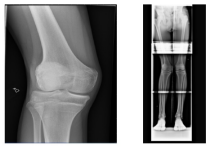
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Guest lecture in orthopedic surgery

Karolinska Institutet

Distal femoral osteotomy

- Increasing valgus



Orthopedic surgery



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Guest lecture in sports and foot/ankle biomechanics

GIH THE SWEDISH SCHOOL OF SPORT AND HEALTH SCIENCES



Healthy subjects
9 intracortical pins:

tibia
talus
calcaneus
navicular
med. cune
metatarsal
phalanx I

cuboid
metatarsal



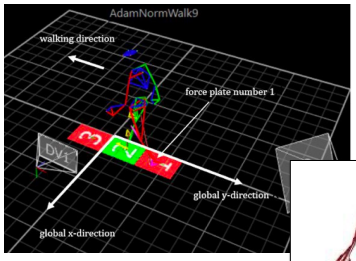
Forefoot compression | Heel compression | Tensile bending stiffness



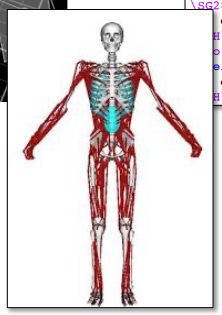
Collaboration with Lanie Gutierrez Farewik and Stefan Hallström, KTH, Stockholm and World Athletics

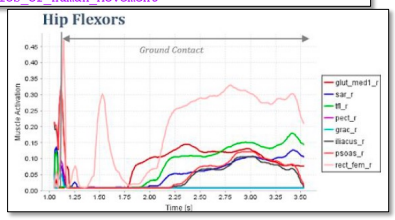
48

Experiments, Projects, Matlab, OpenSim….




```
if movement == 1
    data = readtable("C:\Users\tugdu\Desktop
\KTH\SG2804_Biomechanics_of_Human_Movement
\Projet1\klaidasNormWalk12.txt"); % copy and paste the path to
 the .txt file in "Space Del" folder (e.g. 'C:\KTH\SG2804\Space Del
\ALGait.txt')
elseif movement == 2
    data = readtable("C:\Users\tugdu\Desktop\KTH
\SG2804_Biomechanics_of_Human_Movement\Projet1\klaidasSquat1.txt");
    data2 = readtable("C:\Users\tugdu\Desktop
H\SG2804_Biomechanics_of_Human_Movement
ojet1\klaidasSquatRightLeg3.txt");
eif movement == 3
 data = readtable("C:\Users\tugdu\Desktop
H\SG2804_Biomechanics_of_Human_Movement
```





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Final project: Muscle weakness consequences, compensation and assistance

Goals: to analyze how muscles' force production and the corresponding effort/cost during normal walking are affected by decreased muscle strength, by increased synergist muscle strength and by a passive assistive device.

- Part 1: Introduction of the problem – a weak muscle (soleus)

Part 2 & 3: Different ways to compensate for the weak muscle:

- Part 2: Strengthening an synergist muscle
- Part 3: Use of an external assistive exoskeleton

Analysis of metabolic cost and efficiency, discussion of limitations and realistic implementation

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Final project

- I will finalize the groups this evening.
- Description and files were uploaded today. Note: use new OpenSim model files
- To be presented orally (live or zoom) only
  - no report
  - send slides as *color* pdfs at least 4 business hours before your presentation
- Register for ONE time on Canvas calendar.
- Jenny has some office hours, but she can also meet students at other reasonable times upon agreement

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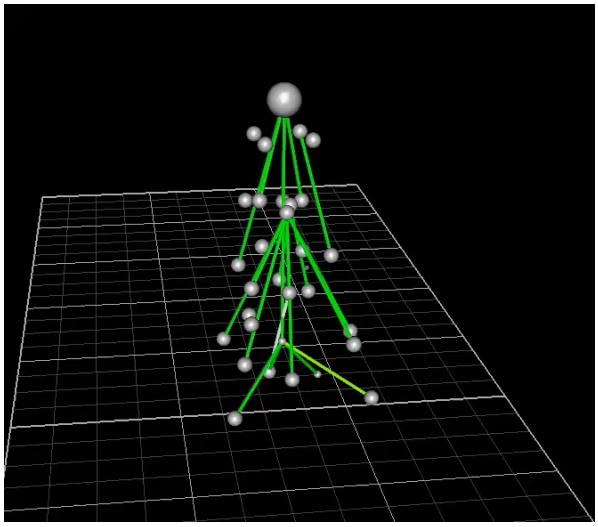
Summary

- Movements, positions and forces in and external to the body'
  - Structure, movement and actuation from muscles, bones and soft tissues
  - Control from CNS
  - Sensory feedback and feedforward
  - Consequences of injury or deviating function in any of above
- Example of amazing engineering!

  But complex! Human behavior is much more complicated than machines
- Knowledge of underlying muscle coordinations and forces is necessary for product design, technique improvements, treatment suggestions, etc.
- Great potential for societal impact when engineers, scientists, innovators, and developers work together with medical professionals!

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Happy holidays!



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